Practical 9th

Considera disk queue with requests f orl/O to blocks on cylinders 98, I83, 4t, 122, t4,
LZ4, 65,67. The SSTF scheduling algorithm is used. The head is initially at cylinder
number 53. The cytinders are numbered from 0 to 199. Write a Program to calculate
the totat head movement (in number of cylinders) incurred while servicing these
requests.

CODE :

```
#include<stdio.h>
#include<stdlib.h>
int main()
{
   int request[8] = {98,183,41,122,14,124,65,67};
   int visited[8] = {0};
   int head = 53;
   int total = 0;
   int i,j;
   printf("Head Movement Sequence:\n");
   for(i=0;i<8;i++)
   {
      int index=-1;
      int min=1000;

      for(j=0;j<8;j++)
      {
         if(visited[j]==0)
         {
            int distance = abs(head - request[j]);
            if(distance < min)
            {
               min = distance;
```

```
                    index = j;
                }
            }
        }
        visited[index] = 1;
        total += min;
        head = request[index];
        printf("%d ", head);
    }
    printf("\nTotal Head Movement = %d\n", total);
    return 0;
}
```

OUTPUT :